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Artificial Intelligence

Bottlenecks Remain for Scaling Physical AI, But Commercialization Progressing

CITI'S TAKE

Citi's 4th annual Robotics and Physical AI Leadership Conference brought together leading founders, investors, and operators at the frontier of physical AI to assess where the market stands and the road ahead. The dominant theme was that commercialization has arrived, but scaling remains hard. Participants distinguished between the promise of physical AI models and the operational reality of deploying them in unstructured, safety-critical environments. Labor shortages, reshoring, and favorable regulatory tailwinds are accelerating enterprise demand, while data scarcity, talent constraints, battery limitations, and high deployment costs remain key friction points. Humanoids are generating significant investment enthusiasm, but near-term ROI is being driven by purpose-built AMRs and specialized systems from companies like Locus Robotics and Dexterity. The conference reinforced our view that physical AI is a decade-long buildout, with durable value accruing to companies that own the data flywheel, solve real deployment problems, and meet the highest safety standards.

Industrial Automation — Advances in physical AI are expanding addressable markets for automation solutions into new end markets and use cases. We see a durable long-term growth tailwind for automation-exposed industrial companies such as Buy-rated ROK, EMR, HON, SYM, RAL, and BDC as a constrained labor market and accelerating domestic manufacturing activity accelerates demand for physical AI solutions.

Commercialization — Real deployments are generating measurable ROI, with humanoid robots pricing at a discount to human labor and delivering payback periods of approximately one year, Dexterity approaching \$300 million in ARR after tripling revenue for three consecutive years, and Gatik holding \$600 million in contracted autonomous trucking revenue. Data flywheel advantages and RaaS business models are emerging as the key commercial moats separating leaders from the field.

Bottlenecks — Commercial deployments are scaling across logistics, warehousing, automotive, and defense, with Locus Robotics processing 200 units per second across 600 sites and Reliable Robotics holding a \$1 billion aviation automation backlog, though data scarcity, power constraints, and high system integration costs remain the key gating factors, creating durable competitive advantages for companies that have solved these bottlenecks at scale.

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Key AI Takeaways

Physical AI is transitioning from proof-of-concept to commercial deployment, but the path to scale is more operationally intensive than the digital AI analogy suggests. Unlike large language models, where a base model carries a lot of the value, physical AI places the premium on proprietary, task-specific data collected in real-world environments, purpose-built hardware, and safety certification.

Across sessions, participants consistently identified data scarcity as the binding constraint, with Instawork noting that even tens of millions of hours of data being collected in 2026 likely represents only basis points, not percentage points, of what is ultimately needed to achieve high-level robotic performance. Power, battery longevity, and chip architecture are also emerging as critical bottlenecks, with panelists noting that existing semiconductor platforms were designed for datacenter workloads, not real-time edge inference on mobile platforms.

The most commercially advanced companies, whether in humanoids, warehouse AMRs, autonomous trucking, or construction, shared a common profile: they started with a specific, high-pain labor problem, adopted a Robotics-as-a-Service model to lower customer adoption barriers, and prioritized safety and reliability above model sophistication.

Key Industrials Takeaways

After attending Citi's Robotics & Physical AI Leadership Conference, we came away convinced that automation, robotics, and physical AI continue to make gradual progress toward broader commercialization, creating what we view as a durable long-term growth tailwind for our automation-exposed industrial companies. Our preferred ways to gain exposure to industrial automation are through Buy-rated ROK, EMR, and HON (pure-play automation providers), SYM (warehouse automation), RAL (sensors and T&M), and BDC (industrial networking), as we view these companies as well positioned to benefit from increasing investments in automation (including equipment, software, and AI). Key drivers of automation adoption remain a constrained labor market as well as accelerating domestic manufacturing activity and capacity expansions, with automation supportive of higher throughput, increased uptime, and improved operational efficiency/accuracy that appears supportive of healthy ROIs.

While logistics, warehousing, and autos appear to be important end markets driving automation adoption (higher volume, repetitive tasks), several panelists highlighted AI's potential to unlock new skills/capabilities for robotics and automation, which could, over time, expand addressable markets for automation solutions (new end markets and new use cases), which we view as a broad positive. Advances in AI/LLMs along with growing availability of data (real-world as well as simulated) are leading to more refined technology through, for instance more integrated hardware and software and more active usage driving increasingly capable deployments (as AI-enabled systems grow "smarter"), which we think could further support accelerating automation adoption over time. In some instances, access to high-quality data remains a bottleneck (where companies are combining simulations with limited real-world data), but we also view this as an opportunity and competitive moat for our companies, given their large installed base and ability to leverage/mine data to further drive autonomy and operational efficiency.

Select companies also highlighted robotics-as-a-service (RaaS) business models that potentially lower the barriers to adoption for small and medium sized

enterprises given lower or minimal upfront capital costs, which we view as supportive of our constructive view on SYM's warehouse-as-a-service offering (GreenBox/Exol), which we think could help drive increasing adoption of SYM's warehouse automation solutions amongst a broader range of customers.

Session Summaries

Investor Panel: Robotics and Physical AI as the Next AI Frontier for Investors

Speakers: Aaron Goldman, Managing Director, General Atlantic; Bill Boyd, Chief Strategy Officer of Symbiotic; Isaac Schecter, Head of Private Capital Markets, Americas at Millennium Management

Approximately \$20 billion has been deployed into physical AI over the past 24 months, though panelists were candid that valuations appear stretched relative to near-term fundamentals. Commercialization was identified as the true pivot point, with Symbiotic advising companies to wait for genuine proof points before raising, keep KPIs simple, and recognize that physical AI requires significantly more operational handholding than software-first businesses.

On foundation models and commoditization, panelists noted that the transformer architecture and pre-training and post-training consensus that exists for large language models has not yet emerged for the models that control robots, where Vision-Language-Action (VLA) models, world models, and various combinations remain in active debate. Data remains in short supply precisely because the industry has not yet converged on what kind of data it needs, and the hardware-software coupling in physical AI is far tighter than in the digital AI stack. Symbiotic's focus centers on filling portfolio gaps, extending battery longevity, and identifying what is genuinely solving a revenue-generating problem at scale.

Pre-IPO appetite for physical AI is expanding, with public markets prioritizing companies that can demonstrate scale, long-term contracts with major retail or logistics customers, and differentiated technology roadmaps with anchor customers 6 to 12 months out. The US and China are identified as the dominant innovation centers, with China holding a meaningful advantage in hardware manufacturing, and price points on humanoid robots approaching \$20,000.

How Robotics Companies Evolve in the Era of Physical AI

Speakers: John Ha, CEO of Bear Robotics; Sean Hsu, CEO of Botrista; and Rick Faulk, CEO of Locus Robotics

Each panelist described a scaled commercial deployment achieved through a differentiated go-to-market approach. Locus Robotics, which has built the largest RaaS business in the US with around 600 sites globally, and 200 units processed per second across its network, succeeded by targeting brownfield warehouse buildings with a demonstrable ROI story rooted in the logistics and 3PL DNA of its founding team.

Bear Robotics has deployed 16,000 robots across 5,000 active sites in hospitality, hotels, casinos, factories, and warehouses, achieving scale by defining its core market across three geographies simultaneously and leveraging strategic investors

to accelerate international sales, dramatically reducing total cost of ownership by solving more than 90 percent of issues remotely without deploying field engineers.

Botrista's approach centered on a six-week payback period for customers and near-zero enterprise churn, achieved by bundling consumables with its RaaS beverage solution into a single sticky package for more than 6,000 locations across F&B, hospitality, schools and office spaces.

Panelists identified the next wave of progress as agentic coordination at scale. Bear Robotics has built an agentic fleet management system from day one, allowing robots to communicate peer-to-peer and build consensus to maintain resilience against network failures, a critical design choice given that networking failures are the number one cause of field failures in dense deployments.

Locus's newest product, Array, uses a visual AI stack that learns with every pick to maximize efficiency and accuracy across an installed base approaching eight billion picks, building a data network effect across different robot form factors. On the five-year outlook, Bear Robotics expects physical AI to enable robots to understand human environments through gesture and voice commands at a higher level and Locus envisions fully lights-out dark warehouses operating 24 hours a day. Panelists agreed that the challenge today is no longer whether robots work but change management on the human side.

Fireside Chat: Velaura AI

Speaker: Rajiv Khemani, CEO of Velaura AI

Khemani argued that the physical AI world requires a complete semiconductor redesign from the ground up. The technology stack for robotics has evolved across three eras: pure CPU-based software algorithms for fixed-function tasks, CNN-based visual perception models, and the current transformer-based generation, but none of these architectures were purpose-built for the unique requirements of robotics at the edge.

Purpose-built chips for physical AI must prioritize ultra-low power (to obtain maximum performance from humanoid robots while avoiding overheating), real-time inference without server roundtrips, and integrated safety modules capable of making decisions locally with minimal false positives. Velaura's technology is already deployed across more than 30 million chips in the field through its work with hyperscalers, providing a proven foundation for its physical AI ambitions.

Khemani identified the key bottleneck areas for scaling physical AI beyond compute: sensor innovation (partnering with RealSense in this area); memory; and battery technologies. The competitive dynamic mirrors the broader AI chip market, where incumbent players designed for very large, homogeneous markets struggle to serve the segmented, specialized requirements of physical AI applications, creating a structural opening for purpose-built entrants.

On defense, Khemani noted that throwaway drones currently achieve accuracy rates of 50 to 55 percent and improving that to 80 to 90 percent constitutes a national security-grade advantage, making defense one of the highest-value near-term applications for purpose-built physical AI silicon.

Physical AI Building the Physical World: Robotics in Construction and Manufacturing

Speakers: Rodion Shishkov, CEO of AI13; Noah Ready-Campbell, CEO of Built Robotics; and Edward Mehr, CEO of Machina Labs

Panelists described three distinct applications of physical AI in the built environment. Built Robotics retrofits existing heavy construction equipment from manufacturers such as Caterpillar with custom software and models, enabling autonomous operation primarily in energy and solar construction, which accounts for approximately 80 percent of new power generation projects today.

Machina Labs operates general-purpose robotic platforms called RoboCraftsman for metal structure fabrication across aerospace, defense, and automotive sectors, completing different parts in a single day across a distributed factory model. AI13 is applying robotics to multi-story residential construction, a domain characterized by almost entirely unstructured environments.

All three companies identified data generation as a primary challenge and strategic asset. Machina Labs noted that its high-rate data collection now enables customers to perform quality control in entirely new ways and generates high-bandwidth physical data streams capable of producing new scientific discoveries. AI13 sees its data layer as a platform for third-party developers to build on, while Built Robotics views operational data from autonomous solar construction as an additional value proposition to accelerate customer adoption. On the question of robotics market hype, Noah Ready-Campbell expressed the view that humanoids are likely overhyped in construction specifically, where specialized equipment that becomes progressively smarter over time is more practical. Edward Mehr argued that the manufacturing conversation needs to be more strategic, with flexibility and distributed manufacturing as the critical advantages rather than replication of existing China-style mass production.

The three biggest scaling challenges identified across panelists were qualification burdens in manufacturing, particularly for safety-critical components in aerospace and automotive; manufacturing execution precision, where small errors in hardware fabrication become deployment blockers; and talent scarcity, with AI13 employing approximately 50 roboticists speaking 10 to 15 languages to address the global nature of the problem. Regulatory standards development was also flagged as a significant gap.

General-Purpose Physical AI Models and Robotics Shaping the Physical World

Speakers: Samir Menon, CEO of Dexterity and Lindon Gao, CEO of Dyna Robotics

Dexterity has built general-purpose AI models enabling hardware-agnostic robots to perform human-level skills at customers including FedEx across e-commerce, parcel, and airport applications, and is approaching \$300 million in ARR after tripling revenue year over year for three consecutive years. Dyna Robotics, backed by \$150 million from Nvidia and other corporate venture investors, builds general-purpose robots for commercial deployment.

Both panelists described a three-layer model architecture for physical AI: an observation layer for understanding physical and semantic space, a reasoning layer, and an actioning layer that translates instructions into motor commands.

General-purpose physical AI is exciting precisely because scaling laws apply: as data produced through reinforcement learning accumulates, robots become dramatically better, crowdsourcing intelligence from hundreds of thousands of training interactions.

On data and commercialization, Dexterity identified expert tokens, defined as optimal responses to unpredicted, unforeseen real-world events in production, as the highest-quality data and the true source of model scale, and expects the industry to remain data-starved for the foreseeable future. Dyna Robotics cautions against smaller specialized models, having found they do not scale well, and instead emphasizes that a large generalized pre-trained model provides the physical grounding and language-to-action translation capability that enables post-trained consumer applications.

On monetization, Dexterity operates a subscription model that it views as more lucrative than token consumption, while Dyna Robotics expects the industry to converge on a token-to-output model over the long term. The panel agreed that physical AI models still need to reach the sophistication level where goals can be issued on the fly, as is already possible in digital AI, and that reasoning models requiring substantial production data to train will take approximately three years in deployment to develop meaningfully.

IBM Presents: Turning Visual Inspection into Actionable Asset Intelligence

Speaker: Ed Neubecker, Principal Automation Technical Specialist at IBM

IBM has brought computer vision into asset management with a platform that enables domain experts to build and own AI models for use at the edge. A 1 percent improvement in quality translates to millions to hundreds of millions of dollars of value for manufacturing operators, and IBM's platform is in active use at Ford for F150 truck inspection and at Sund and Baelt Bridge using Maximo to reduce maintenance risk.

The addressable asset maintenance opportunity spans bridges, generators, HVAC systems, utilities, and IT datacenter infrastructure. The platform supports 2D vision across thermal, x-ray, and standard imagery including drone video feeds, with 3D inspection in development. IBM's build-your-own model design ensures clients retain full data ownership, a key differentiator for enterprises where visual inspection processes represent proprietary operational IP.

How AI Is Changing the Way We Deploy Automation in Factories and Warehouses

Speakers: Ross Diankov, CEO of Mujin; and Etienne Lacroix, CEO of Vention

Mujin was founded 15 years ago with the mission of deploying as many industrial robots as possible, now exceeding four-digit deployments globally, with a focus on providing whole turnkey automation systems rather than individual component technologies. Vention is solving the deployment and system integration problem in physical AI and robotics, operating across approximately 4,000 factories with 30,000 pieces of equipment in the field, primarily serving Fortune 500 companies.

Both panelists offered a candid assessment of where physical AI sits on the maturity curve: there are plenty of industrial-grade-ready applications from a

reliability and cycle-time perspective, but commercial readiness is roughly at the ChatGPT 2.5 equivalent, not ChatGPT 3, because the marginal cost of distribution for physical AI remains very high versus zero for LLMs. Everything in physical AI requires wiring, system integrations, commissioning, and engineering work, and the cost of failure is categorically different from digital AI. ROI expectations from physical AI customers center on a 1-to-2-year payback period, and a hidden challenge for many physical AI companies is the cost of the data gathering process, which can run into tens of millions of dollars.

On macro tailwinds, both panelists noted meaningful acceleration in demand since the fourth quarter of 2025. Vention observed the PMI inflection before public data confirmed it, attributing the shift to reshoring-driven investment in local manufacturing, immigration policy effects on labor pool availability, and enterprise demand to adopt AI. Mujin pointed to reshoring as a fundamental driver, noting that the mindset of robotics as an investment in competitive survival, rather than a cost-reduction tool, is now filtering into American businesses. Mujin uses simulation as a continuous safety line, running hourly regression tests in simulation and validating real-world robot behavior against simulated results. Vention's strategic focus has been on shrinking system integration costs from 40 percent of project value to 10 to 15 percent, with value in physical AI residing in the data flywheel rather than in the VLA model layer itself.

Autonomy and Mobility: Ground vs. Air

Speakers: Gautam Narang, CEO of Gatik and Mark Stoll, CFO of Reliable Robotics

Gatik is the only player conducting truly driverless operations on public roads across 29 states for customers including Pepsi, Coke, Walmart, Kroger, and Tyson with \$600 million in contracted revenue. Narang described the industry as somewhere between the scaling phase, where economics are being proven, and the physical AI as a business phase, with mainstream commercial deployment of autonomous trucking expected before year-end 2026 and hundreds of trucks commercially deployed.

Reliable Robotics is building enhanced systems to fully automate existing fixed-wing aircraft, with a fundamentally different regulatory profile: unlike ground vehicles, aviation requires proving safety before operating, not after. Reliable is the only company to have achieved FAA means of compliance sign-off on its certification approach, which Stoll identified as a critical and proprietary moat, a skeleton key that can be applied across different aircraft types. On the regulatory landscape, Narang noted that the US regulatory framework for autonomous trucking is highly permissive and that the current administration is expected to roll out a national framework soon.

Reliable Robotics is targeting full automation delivery by end of 2027 and is currently executing on a \$17 million Air Force contract in Guam along with a special operations command contract. Reliable's commercial backlog stands at one billion dollars, with its current bottleneck being talent for safety-critical engineering. Stoll predicted that autonomous trucking will have a shorter adoption curve than aviation given its more established regulatory path, but that once autonomous aircraft certification is achieved, the logistics implications are transformational: airports could function like bus terminals, a ~\$4.5 trillion GDP aviation sector could accommodate 10x the aircraft (with military use-cases leading logistics and

commercial aviation adoption), and the certification engine would be defensible for 20 to 40 years given the difficulty of ripping out certified avionics systems.

Breaking the Data Bottleneck in Physical AI

Speakers: Sumir Meghani, CEO of Instawork; Nadav Orbach, CEO of RealSense; and Simon Box, CEO of ReSim.ai

Panelists described the demand for physical AI training data as growing exponentially: in 2024, the largest labs were generating hundreds of thousands of hours of data; in 2025, millions of hours; and in 2026, tens of millions of hours. Even at this scale, Instawork characterized the current collection as representing basis points, not percentage points, of the data needed to achieve high-level robotic performance, underscoring that LLMs are not a valid comparison because they were able to bootstrap from all human knowledge on the web.

On data types and quality, panelists identified teleoperation data as extremely valuable but difficult to scale, and synthetic simulation data as abundant but not yet sufficiently useful across all domains, with RealSense describing a working mix of approximately 60 percent simulated to 40 percent real-world data in some scenarios. The panel's primary bottleneck identification was evaluation, not data generation: determining whether a newly trained model is ready for deployment requires running millions of simulations and real-world tests, and events or issues that robots encounter that were not captured in the simulation set remain the most common complaint. ReSim.ai's business model is usage-based, while Instawork is monetizing through data collection services and anticipates a growing service economy around certified robot maintenance technicians.

Companies Mentioned:

Belden Inc (BDC.N; US\$107.41; 1; 07 Jul 26; 16:00)

Emerson Electric Co. (EMR.N; US\$137.91; 1; 07 Jul 26; 16:00)

Honeywell International Inc (HON.O; US\$225.05; 1; 07 Jul 26; 16:00)

Ralliant Corp (RAL.N; US\$67.75; 1; 07 Jul 26; 16:00)

Rockwell Automation (ROK.N; US\$468.9; 1; 07 Jul 26; 16:00)

Symbolic Inc (SYM.O; US\$41.32; 1H; 07 Jul 26; 16:00)

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